

Solution

LIGHT-03

Class 08 - Science

1.
(c) Angle of reflection
Explanation: The Law of Reflection states that the angle of the incident light ray is equal to the angle of the reflected light ray.
2.
(a) Converging of light rays.
Explanation: Converging of light rays.
3.
(c) Images of Sun
Explanation: The small patches of sunlight under the tree are the image of sun formed by small gaps between tree leaves.
4.
(c) it does not reflect any colour
Explanation: it does not reflect any colour
5.
(c) 160°
Explanation: 160°
6.
(b) Same size and erect
Explanation: Plane mirror always forms virtual, erect, image of same size as object. So, when you stand in front of small mirror image will be of same size.
7.
(a) Due to reflection
Explanation: The reason why the air bubble shines in water is because of the reflection. The speed of light inside the air bubble is faster than in the water. The reason why the air bubble appears to be like a mirror is that the light gets reflected around the outside of the bubble.
8.
(c) Concave mirror
Explanation: Concave mirror
9.
(a) Scatter least
Explanation: Danger signals are made red coloured because red light have longer wavelength that scatter less. Red coloured light can be seen from distance due to less scattering.
10.
(a) Refraction of light
Explanation: Refraction of light
11.
(d) Cinematography
Explanation: Cinematography is based on the principle of persistence of vision. Colour blindness, hemophilia, and myopia are defects of vision.
12.
(b) Rays of light
Explanation: A light ray is a line (straight or curved) that is perpendicular to the light's wavefronts; its tangent is collinear with the wave vector
13.
(b) spherical
Explanation: spherical
14.
(a) diverging lens
Explanation: diverging lens

15. **(b)** Image formed by plane mirror
Explanation: The image formed by a plane mirror is always virtual, erect, and of the same size as the object. Image formed on retina, pinhole camera, and cinema screen are real images.
16. **(a)** (i), (ii) and (iii)
Explanation: The images by plane mirror are
Image distance = object distance
Image size = object size
Laterally inverted = Erected image
17. **(a)** Convex
Explanation: Convex
18. **(a)** Between focus and centre of curvature
Explanation: Between focus and centre of curvature
19. **(b)** Convex mirror
Explanation: Convex mirror is used as shop security mirror because it covers a larger area and forms an erect image of the object to view a larger section of the shop.
20. **(d)** iris
Explanation: The iris is the area of the eye that contains the pigment which gives the eye its colour,
21. **(d)** Silver
Explanation: Silver
22. **(a)** Focal length
Explanation: Focal length
23. **(a)** Always
Explanation: The angle of incidence is always equal to the angle of reflection of light. If the angle between mirror and incident ray is 90° then the angle of reflection will be 90° .
24. **(c)** 50°
Explanation: The angle of incidence is always equal to the angle of reflection of light. The angle between mirror and incident ray is 50° then the angle of reflection will be 50° .
25. **(c)** Real, inverted, diminished
Explanation: Real, inverted, diminished
26. **(d)** inverted, real, smaller
Explanation: inverted, real, smaller
27. **(d)** real, erect and enlarged
Explanation: real, erect and enlarged
28. **(d)** Persistence of vision
Explanation: Videography and cinematography is based on principle of persistence of vision in which image formed on retina remain for $1/16$ th second. So, picture moving at the speed with 24 frames in a second appear to be connected with each other.
29. **(d)** Do not require material medium for propagation
Explanation: Light consists of electromagnetic waves that do not require material medium for its propagation. It can travel in vacuum.

30. **(a)** Real images
Explanation: Real images
31. **(a)** Isaac Newton
Explanation: Isaac Newton
32. **(c)** retina
Explanation: retina
33. **(d)** bends towards the normal
Explanation: bends towards the normal
34. **(c)** reflection
Explanation: reflection
35. **(a)** infinite
Explanation: A ray of light can be reflected infinite times by two plane mirrors placed parallel and facing each other because each reflected ray would be the incident ray for the other mirror.
36. **(d)** does not obey the laws of reflection
Explanation: Diffused reflection of light does not obeys the law of reflection of light as reflected rays are not parallel to each other.
37. **(a)** the world would become two-dimensional for humans
Explanation: the world would become two-dimensional for humans
38. **(c)** Hypermetropia
Explanation: A person suffering from Hypermetropia or farsightedness is not able to view the nearby object clearly but can see a far object distinctly. It can be corrected using a convex lens of proper power.
39. **(b)** regular reflection
Explanation: Image is formed by regular reflection. When a ray of light approaches a smooth polished surface and the light ray bounces back, it is called the reflection of light.
40. **(a)** Dispersion
Explanation: The phenomenon of splitting of white light into seven colours is called dispersion of light to obtain spectrum. These colors are often observed as light passes through a triangular prism. Upon passage through the prism, the white light is separated into its component colors - red, orange, yellow, green, blue, and violet.
41. **(b)** Volt
Explanation: Volt
42. **(a)** light travels in straight line.
Explanation: light travels in straight line.
43. **(b)** Refraction of light
Explanation: The phenomenon of bending of rays of light when passes from one medium to another medium where its speed is different is called refraction of light. In Refraction of light when it passes from a rarer medium to a denser medium it bends towards the normal to the boundary between the two mediums.
44. **(a)** 25 cm to infinity
Explanation: For a normal human eye, the range of vision is between infinity and 25 cm.
45. **(d)** Colour blindness

Explanation: Colour blindness is the defect of vision due to absence of cones in retina. Cones are responsible to see the colour of the object. Some people do not contain certain cones which enable them to see colour.

46. (a) Light

Explanation: Light is the physical quantity that do not require medium for propagation. Sound require medium for propagation. Light can travels in vacuum also.

47. (a) Violet colour

Explanation: Violet colour

48.

(b) biconvex

Explanation: biconvex

49.

(d) 5 m

Explanation: 5 m

50.

(c) 45°

Explanation: The angle of incidence is always equal to the angle of reflection of light. If the angle between mirror and incident ray is 45° than angle of reflection will be 45° .

51.

(b) Concave or convex

Explanation: The convex and concave lenses are used to correct the defective vision. Myopia can be corrected by placing a suitable concave lens in the line of sight. Hypermetropia can be corrected by placing a suitable convex lens in the line of sight. To correct presbyopia, a bifocal lens is used.

52.

(c) The near point has receded away

Explanation: The near point has receded away

53.

(c) Perpendicular to surface

Explanation: Perpendicular at the point of incident of light to the surface is called normal. Normal is an imaginary line.

54.

(b) Green & Red

Explanation: Most people with color blindness can't distinguish between certain shades of red and green. Less commonly, people with color blindness can't distinguish between shades of blue and yellow. Certain eye diseases and some medications also can cause color blindness.

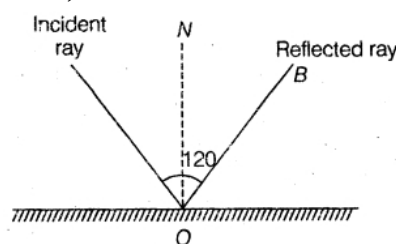
55.

(d) Presence of dispersing medium

Explanation: The path of the light is visible in presence of dispersing medium like dust particle or prism or other substances in the medium.

56. (a) 60°

Explanation: According to the question,
Given,



$$\angle AON + \angle NOB = 120^\circ$$

According to law of reflection,

Angle of incidence = Angle of reflection

Therefore, $\angle AON = \angle NOB$

$$\angle AON = \frac{120}{2} = 60^\circ$$

Therefore, the angle of incidence = 60°

57.

(c) virtual and smaller

Explanation: virtual and smaller

58.

(c) behind the retina

Explanation: behind the retina

59.

(c) 5m

Explanation: 5m

60.

(c) Vacuum

Explanation: Vacuum

61.

(b) At a distance less than its focal length.

Explanation: At a distance less than its focal length.

62.

(b) Blue light

Explanation: Blue light has shorter waves, with wavelengths between about 450 and 495 nanometers. Red light has longer waves, with wavelengths around 620 to 750 nm. Blue light has a higher frequency and carries more energy than red light.

63.

(a) total internal reflection of light

Explanation: total internal reflection of light

64.

(a) Reflection of light

Explanation: Reflection of light enables us to see our face in mirror as mirror forms image behind the mirror which is virtual and erect.

65.

(d) Statement D is correct.

Explanation: Both Convex lens is thicker in the middle and thinner at the edges and Convex lens is a converging lens

66.

(c) Vitamin A

Explanation: Vitamin A plays a crucial role in vision by maintaining a clear cornea, which is the outside covering of your eye.

67.

(a) $\frac{1}{F}$

Explanation: $\frac{1}{F}$

68.

(d) they are emitting their own light

Explanation: they are emitting their own light

69.

(a) Focal length

Explanation: Focal length

70.

(c) Concave mirror

Explanation: Concave mirror

71.

(b) Large number of image are formed

Explanation: When two mirrors are placed parallel to each other, large numbers of images are formed due to repeated reflection from opposite side placed mirror.

72.

(a) Magnified and erect

Explanation: Magnified and erect

73.

(c) 60 degrees

Explanation: The ray of light strikes the mirror at an angle of 30° , which means it will make an angle of 60° ($90^\circ - 30^\circ$) with the normal.

So the angle of incidence = 60°

According to the law of reflection, the angle of incidence is equal to the angle of reflection.

Angle of reflection = 60°

74. **(a)** Cornea

Explanation: Eye donation involves donating corneas. Sometimes it called a keratoplasty, or a corneal graft, a cornea transplant could give someone back the gift of sight.

75. **(a)** Red, Green, Blue

Explanation: Red, Green, Blue